Tissues

The big idea

In multicellular organisms, cells that are similar in structure and work together for a particular function form a **tissue**. Plants and animals have different tissues because of their different lifestyles: plants are fixed, animals move.

Plant tissues

Meristematic tissue (dividing cells; thin walls, dense cytoplasm, prominent nucleus, no vacuoles):

- **Apical** at root and shoot tips (length), **lateral** (cambium; girth), **intercalary** at the base of leaves and internodes.

Permanent tissues (cells that have lost the ability to divide):

Tissue	Features	Role
Parenchyma	Loosely packed living cells, thin walls	Storage; **chlorenchyma** (photosynthesis), **aerenchyma** (buoyancy in water plants)
Collenchyma	Walls thickened at corners	Flexibility (bending of leaves and stems)
Sclerenchyma	Dead cells with **lignin**	Hardness (husk of coconut)
Epidermis	Outer layer, waxy cutin; **stomata**	Protection, gas exchange, transpiration
Cork	Dead cells with **suberin**	Protection in older stems

Complex permanent tissues (more than one type of cell):

- **Xylem**: tracheids, vessels, xylem parenchyma, xylem fibres; carries **water and minerals** upward.
- **Phloem**: sieve tubes, companion cells, phloem parenchyma and fibres; carries **food** in both directions.

Animal tissues

Tissue	Types and roles
Epithelial	Covers and lines organs: squamous (skin, lining of mouth), cuboidal (kidney tubules), columnar (intestine), ciliated (respiratory tract), glandular
Connective	Blood (plasma + RBC, WBC, platelets), bone (hard matrix of calcium and phosphorus), cartilage, ligament (bone to bone), **tendon** (muscle to bone), areolar, adipose (fat storage)
Muscular	**Striated** (voluntary, skeletal), **smooth** (involuntary, in the gut), **cardiac** (involuntary, heart; never tires)
Nervous	**Neurons**: cell body, dendrites, long **axon**; carry impulses quickly

LIGAMENT VS TENDON

Ligament joins bone to bone (think "link"). **T**endon joins muscle to bone (think "tug").

Quick recap

- Plants: meristematic (dividing) and permanent tissues; xylem and phloem conduct.
- Animals: epithelial, connective, muscular, nervous.
- Know where each tissue is found and what it does.

Practice questions

1. Growth in length of the stem is due to
   (a) apical meristem (b) lateral meristem (c) intercalary meristem only (d) cork
2. The husk of a coconut is made of
   (a) parenchyma (b) collenchyma (c) sclerenchyma (d) chlorenchyma
3. Which tissue carries food in plants?
   (a) Xylem (b) Phloem (c) Epidermis (d) Cork
4. A tendon connects
   (a) bone to bone (b) muscle to bone (c) muscle to muscle (d) nerve to muscle
5. Cardiac muscle is found in the
   (a) arms (b) stomach (c) heart (d) legs

Answers: 1. a 2. c 3. b 4. b 5. c